Design and Analysis of Software Systems

SPRING 2026 - Mid-Semester Exam

INSTRUCTIONS: This is a closed-book exam. You may use a ONE page hand-written cheatsheet.

It consists of 13 questions. You should answer all questions. You have 90 minutes to complete the exam for a total of 45 points.

Credit is given for what you write, not what you are thinking. Partial credit will be given based on Quality of the content, not Quantity. A voluminous content-free answer may simply annoy the grader.

1. **(1 point)** In the context of project scheduling, what is the primary benefit of utilizing the Critical Path Method (CPM)?
   A. It enables flexibility in task management by eliminating critical tasks.
   B. It identifies the longest sequence of tasks to determine project duration, facilitating effective scheduling.
   C. It allows for arbitrary prioritization of tasks regardless of their dependencies.
   D. It emphasizes the importance of individual task completion without considering overall project impact.

2. **(1 point)** In requirements engineering, which of the following techniques is most effective for eliciting high-quality requirements from stakeholders?
   A. Surveys that gather quantitative data.
   B. Joint Application Development sessions that facilitate collaborative discussions among stakeholders.
   C. Reviewing previous projects' documentation without stakeholder involvement.
   D. Solely relying on email communications to gather requirements from stakeholders.

3. **(1 point)** In UML class diagrams, how do multiplicities enhance the understanding of relationships between classes?
   A. They illustrate the visibility levels of class attributes and methods across different classes.
   B. They serve to define the methods and attributes of classes without reflecting their relationships.
   C. They detail the sequence of messages exchanged between objects in a communication diagram.
   D. They specify the exact number of instances from each class that can participate in the relationship, indicating constraints on associations.
   E. None of the above

4. **(1 point)** In unit testing, assertions are used to confirm that the expected outcomes match the actual results. A common assertion method is _____, which checks that a specific condition holds true during the test.
   Assert true

5. **(1 points)** A software team is building an online exam portal for a university. Their requirements document includes the following statement:

   "The system shall ensure that the exam experience is smooth and secure for all users."

   Which TWO quality defects does this requirement violate?

A) Infeasible and Inconsistent
B) Ambiguous and Not Verifiable
C) Incomplete and Infeasible
D) Not Necessary and Not Traceable

6. **(2 points)** A developer is practising TDD on a ShippingService module. She first writes a failing test for STANDARD shipping (RED), then implements the minimum code to pass it (GREEN). Over subsequent iterations, she adds EXPRESS, OVERNIGHT, and INTERNATIONAL shipping types using if-else chains inside a single method, and all tests pass. She concludes the implementation is complete. Her senior engineer disagrees and says the codebase now has technical debt. Is the senior engineer correct? Explain clearly. Refacto

7. **(2 points)** In Bottom-Up Integration Testing, what is the primary reason that drivers are considered cheaper to write than stubs in Top-Down Integration Testing?

8. **(4 points)** A development team at a fintech company has set up the following pipeline: every code push triggers an automated build, unit tests, and integration tests. However, deploying to production requires a senior engineer to manually approve and trigger the release after reviewing the test report. The team argues that skipping Continuous Deployment is justified for a fintech system. Using concepts from the process model lectures (specifically DevOps), give two specific technical or operational reasons why manual approval before production deployment is appropriate in this context.

9. **(4 points)** MedTrack is a critical hospital software system used to manage patient records, medication scheduling, and billing. The development team is 12 engineers working in a hybrid Agile model with 2-week sprints.

   The system has the following key modules:
   - Module A – Patient Registration (UI Layer)
   - Module B – Electronic Health Records (EHR) (Database Layer)
   - Module C – Medication Scheduler (Logic Layer)
   - Module D – Billing & Insurance Gateway (External API Integration)

   During Sprint 7, the team completed Modules A, B, and C. Module D is still under development. A critical bug was recently discovered in production: a patient was administered the wrong medication dose because Module C read an outdated record from Module B when two concurrent update requests occurred.

   The QA lead is now tasked with designing a comprehensive testing strategy before the next release, which will include Module D for the first time.

   The production bug (Module C reading stale data from Module B under concurrent updates) was not caught during testing. Identify the exact testing level(s) that should have caught this defect and explain why each may have been insufficient in this case. What specific test scenario would have exposed this bug?

10. **(6 points)** Develop an automated ordering system for the The Most Amazing Coffee Shop, where customers can place orders at self-service kiosks (SSKs) located inside the shop. Each order will include the customer's selection of drinks (e.g., espresso, latte, cappuccino), size (small, medium, large), any added customizations (e.g., milk type, flavor shots), and payment information. Once an order is placed and payment is confirmed, the system will generate a

receipt, and the customer will be notified when their order is ready for pickup. The coffee shop staff will also be able to update the menu and availability of items in real-time.
Note: You may make simple assumptions about this problem based on your own knowledge of coffee shop ordering systems.

a) **(3 points)** Draw the usecase model for the automated ordering system clearly showing all the usecases, primary and supporting actors.
b) **(3 points)** Write a detailed textual use case (step-by-step) for a customer successfully placing an order using the SSK.

11. (6 points) You are the project manager for a hospital appointment scheduling system. After 6 weeks of a planned 20-week project, you collect the following EVM data:

- **Budget at Completion (BAC):** $200,000
- **Planned Value (PV):** $60,000
- **Earned Value (EV):** $48,000
- **Actual Cost (AC):** $54,000

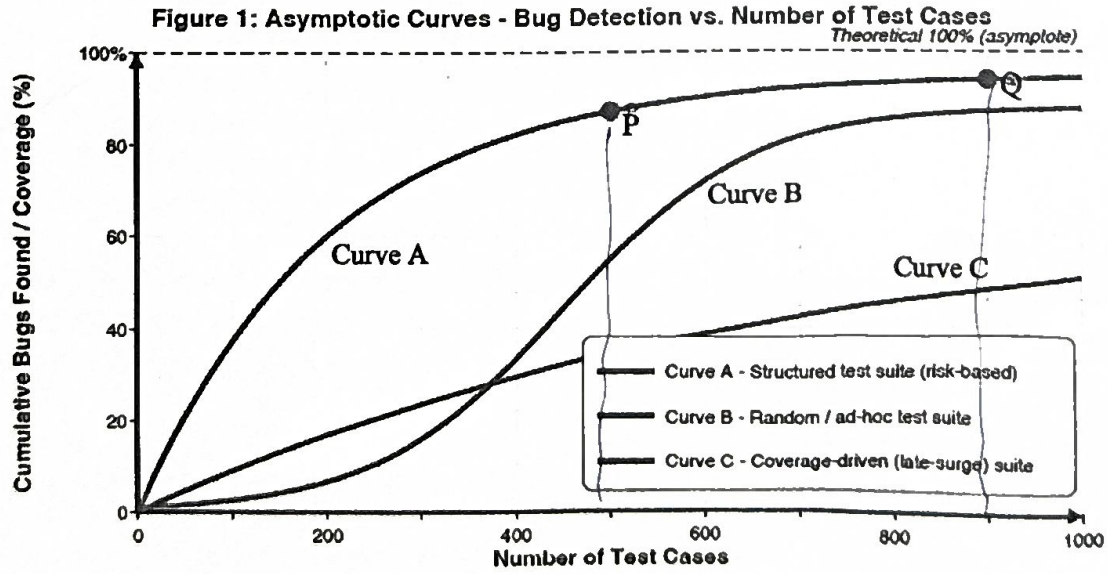
Your sponsor says: *"We've only spent $54K of $200K. We are well under budget, no worries!"*. Your development lead says: *"We finished the login module early, so we're ahead of schedule."*

Tasks:

a. Calculate the Cost Performance Index (CPI) and Schedule Performance Index (SPI) (2 points).
b. Identify what the data actually reveals. Are the sponsor and development lead correct? (2 points)
c. What actions should you take in the next sprint planning session? (2 points)

12. (8 points)
The graph below (Figure 1) shows three asymptotic curves representing cumulative buge detection as a function of number of test cases executed, for three different testing strategies applied to the same software system. The software under test is a hospital patient management system with approximately 800 known defects (as determined post-release via exhaustive analysis). Points P & Q are marked on marked on Curve A at approximately 500 test cases (88% defects found) and 900 test cases (95% defects found) respectively. All three test suites start from zero and test the same code base.

Figure 1: Asymptotic Curves - Bug Detection vs. Number of Test Cases



(a) Curve A approaches but never reaches the 100% line. Explain, using the concept of an asymptote, why no finite test suite can ever guarantee 100% bug detection. **(2 points)**

(b) A junior tester runs the curve A test suite and achieves 95% statement coverage. She concluded: "We have 95% coverage, so we have found 95% of all the bugs". Is the conclusion appropriate? Argue in favor or against the conclusion. **(2 points)**

(c) Identify the testing strategy or approach most likely responsible for curve B and C's shape. Give appropriate reasons. **(4 points)**

13. (8 points) A software project has the following **unadjusted** function point counts:

Type	Low	Medium	High
Inputs	2	3	1
Outputs	1	2	2
Inquiries	4	1	0
Internal Logical Files	1	2	1
External Interface Files	3	1	0

The 14 general system characteristics are each rated as follows: six rated 3, five rated 4, and three rated 2. Assume value adjustment factor as **1.1**. Historical data shows **18 hours of effort per function point**. The team has **4 developers**, each available **30 hours/week**. A **15% buffer** is added to the final timeline.

What is the **total project duration in weeks** (rounded to the nearest whole number)?

Please ensure you show the complete sequence of steps.